

Requirement Specification

E-Commerce Order Management System (OMS)

Version: 1.0

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1. Introduction

1.1 Purpose

This Business Requirement Specification (BRS) defines the functional and non-functional requirements for the proposed E-Commerce Order Management System (OMS). The system aims to centralize the

lifecycle of a digital order from the point of checkout to final delivery and reverse logistics, ensuring a seamless experience for both operators and customers.

1.2 Scope

The system will act as the operational core for all fulfillment activities. It will interface with front-end digital storefronts, back-end ERP systems, payment gateways, and third-party logistics (3PL) providers to automate inventory sync, order routing, picking, shipping, and returns.

2. Functional Requirements

2.1 Inventory Management

Maintaining an accurate, real-time ledger of available stock across all nodes in the fulfillment network is paramount.

Req ID	Description	Priority
FR-INV-01	The system shall synchronize inventory balances across all integrated storefronts (e.g., Shopify, Magento) in near real-time (under 60 seconds delay).	High
FR-INV-02	The system shall calculate 'Available to Promise' (ATP) inventory by deducting reserved/pending orders from physical on-hand counts.	High
FR-INV-03	The system shall support automated alerts when SKU balances fall beneath predefined safety-stock thresholds at any facility.	Medium

2.2 Order Processing & Routing

Orchestrating the journey of an order ensuring validation, fraud checks, and optimal fulfillment sourcing.

Req ID	Description	Priority
FR-ORD-01	The system shall validate delivery addresses and calculate fraud risk scores prior to transitioning an order to 'Accepted' status.	High
FR-ORD-02	The system shall utilize intelligent routing logic to assign order fulfillment to the warehouse or store closest to the customer's delivery address.	High
FR-ORD-03	The system shall automatically execute order splitting if the requested items must be sourced from multiple physical locations.	Medium
FR-ORD-04	The system shall allow customer service agents to place administrative holds, modify line items, or cancel orders before fulfillment processing begins.	High

2.3 Fulfillment & Shipping

Integrating with carriers to facilitate efficient dispatch and visibility.

Req ID	Description	Priority
FR-SHP-01	The system shall integrate with primary carriers (e.g., UPS, FedEx, USPS) via API to generate shipping labels and fetch real-time carrier rates.	High
FR-SHP-02	The system shall auto-generate and transmit digital pick tickets and packing slips to designated warehouse queues.	Medium
FR-SHP-03	The system shall capture tracking numbers upon dispatch and automatically push shipping notification events to the originating storefront.	High

2.4 Reverse Logistics (Returns & Refunds)

Providing structured workflows for managing returned merchandise.

Req ID	Description	Priority
FR-RET-01	The system shall support the generation of Return Merchandise Authorizations (RMA) tied directly to the original order ID.	High
FR-RET-02	The system shall automate the financial refund workflow via the original payment processor once returned inventory is scanned and verified at the facility.	High

3. Non-Functional Requirements

3.1 Performance and Scalability

- **Peak Capacity:** The system architecture must securely handle transaction spikes up to 15,000 orders per hour during major promotional events without degradation in performance.
- **Response Times:** API endpoints responsible for inventory lookups must maintain a P95 latency of under 200 milliseconds.

3.2 Security and Compliance

- **Data Protection:** All Personally Identifiable Information (PII) must be encrypted at rest (AES-256) and in transit (TLS 1.3).
- **Regulatory Compliance:** The environment must adhere to PCI-DSS standards for payment tokenization, as well as GDPR and CCPA guidelines for customer data retention and deletion.
- **Access Management:** Internal user access must be governed by strict Role-Based Access Control (RBAC) and require Multi-Factor Authentication (MFA).

3.3 Reliability and Disaster Recovery

- **Availability:** The platform shall guarantee a 99.99% uptime Service Level Agreement (SLA), achieved via multi-zone cloud redundancy.
- **Recovery:** The system must adhere to a Recovery Point Objective (RPO) of 15 minutes and a Recovery Time Objective (RTO) of 1 hour in disaster scenarios.

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